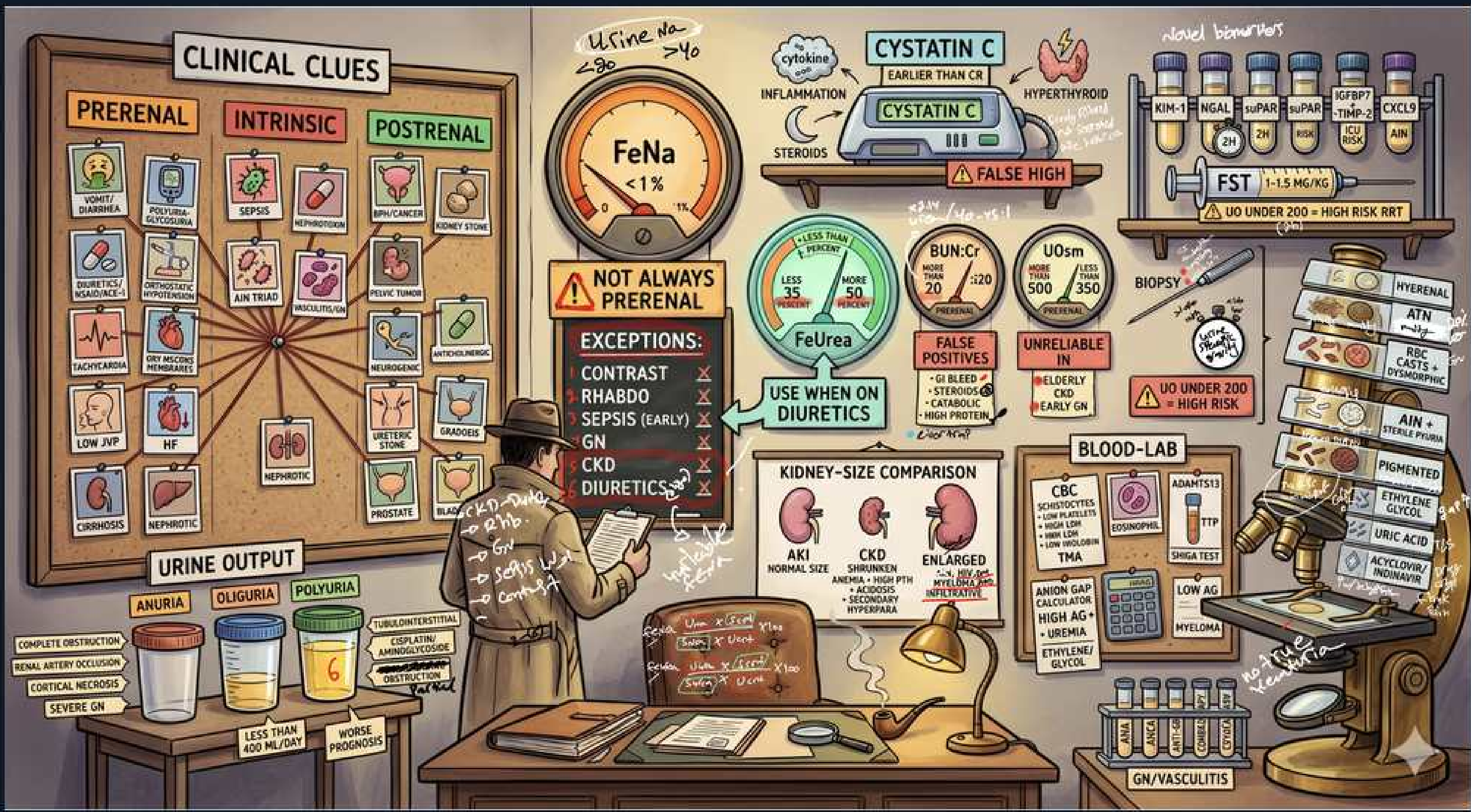
AKI — Diagnosis



1. Clinical clues board

WHAT YOU SEE

CLINICAL CLUES with PRERENAL / INTRINSIC / POSTRENAL columns

WHAT IT ENCODES

The first step in any AKI workup is to classify it into prerenal (hypoperfusion), intrinsic/renal (parenchymal injury), or postrenal (obstruction) using history, exam, urinalysis, and basic labs. Prerenal clues are volume loss, low CO, NSAID/ACE-I use; intrinsic clues are recent contrast/nephrotoxins, casts on micro; postrenal clues are anuria, BPH, single kidney, or distended bladder. Always place a Foley and order a renal ultrasound early to rule out reversible obstruction before chasing intrinsic causes.

BOARD TRAP

Students jump straight to ordering a kidney biopsy or FENa. The correct first move is the cheap triad — bladder scan/Foley, renal ultrasound, and urine microscopy — because postrenal obstruction is instantly reversible and missing it is a classic board error.

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2. FENa <1%

WHAT YOU SEE

Large FENa <1% gauge

WHAT IT ENCODES

FENa = (UNa × PCr)/(PNa × UCr) × 100. A FENa <1% means intact tubules are avidly reabsorbing sodium to defend volume, pointing to PRERENAL azotemia; a FENa >2% indicates tubular cells are too injured to reabsorb Na, pointing to ATN (intrinsic). The 1–2% gray zone is indeterminate. FENa is only valid in oliguric AKI and is invalidated by diuretics.

BOARD TRAP

The trap is treating FENa as a pure prerenal-vs-ATN dichotomy and forgetting it is meaningless if the patient is on a loop diuretic (which dumps Na and falsely raises FENa) — in that setting you must switch to FEUrea.

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3. FENa exceptions

WHAT YOU SEE

NOT ALWAYS PRERENAL exceptions: contrast, rhabdo, sepsis early, GN, CKD

WHAT IT ENCODES

A low FENa (<1%) is NOT exclusive to prerenal disease — it also occurs in contrast-induced nephropathy, rhabdomyolysis/pigment nephropathy, early sepsis, acute glomerulonephritis, hepatorenal syndrome, and CKD, all of which involve Na-avid states or vasoconstriction despite intrinsic injury. Conversely, ATN typically gives FENa >2%. Memory hook: 'CRaSGoCK' lowers FENa even without true prerenal physiology.

BOARD TRAP

Picking 'prerenal' simply because FENa is <1% in a patient with a positive blood dipstick and CK in the thousands — that is rhabdomyolysis with pigment ATN, an intrinsic injury that mimics a prerenal FENa.

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4. FEUrea on diuretics

WHAT YOU SEE

FeUrea circle, 'use when on diuretics'

WHAT IT ENCODES

FEUrea = (UUrea × PCr)/(BUN × UCr) × 100, and it is the test of choice when the patient is on diuretics because urea handling is in the proximal tubule and is unaffected by loop/thiazide action at the distal nephron. FEUrea <35% supports PRERENAL; >50% supports ATN. It rescues you when FENa is uninterpretable.

BOARD TRAP

The classic trap is reporting a 'high' FENa in a patient on furosemide and calling it ATN — the diuretic falsely elevated FENa. The right discriminator on diuretics is FEUrea, not FENa.

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5. BUN:Cr ratio

WHAT YOU SEE

BUN:Cr gauge with false-positive note

WHAT IT ENCODES

A BUN:Cr ratio >20:1 supports PRERENAL azotemia because slow tubular flow lets passive urea reabsorption rise while creatinine (not reabsorbed) stays put. A ratio near 10–15:1 suggests intrinsic ATN where reabsorption is lost. The ratio is confounded upward by GI bleeding, high-protein/TPN intake, corticosteroids, and catabolic states (false-positive prerenal pattern).

BOARD TRAP

Calling a high BUN:Cr 'prerenal' in a patient with melena and an upper GI bleed — the absorbed blood protein raises BUN independent of perfusion, so the ratio is a false positive and the kidneys may be fine.

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6. Urine osmolality

WHAT YOU SEE

UOsm gauge, 'unreliable'

WHAT IT ENCODES

Urine osmolality >500 mOsm/kg with a high specific gravity reflects intact concentrating ability and supports PRERENAL; values approaching serum (~300, isosthenuria) suggest ATN where tubules can no longer concentrate. UOsm is unreliable in CKD, with diuretics, and with osmotic agents (glucose, contrast, mannitol).

BOARD TRAP

Trusting urine osmolality in a CKD patient or one on diuretics — both blunt concentrating ability and make a low UOsm look like ATN when the real process is prerenal.

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7. Urine output

WHAT YOU SEE

URINE OUTPUT row: ANURIA / OLIGURIA / POLYURIA cups

WHAT IT ENCODES

Anuria (<50–100 mL/day) is ominous and points to complete bilateral obstruction, a vascular catastrophe (cortical necrosis, bilateral RAS thrombosis), or severe RPGN. Oliguria (<400–500 mL/day) is common to all AKI types. Importantly, non-oliguric AKI exists — classically aminoglycoside ATN and contrast nephropathy — and carries a better prognosis.

BOARD TRAP

Assuming AKI must be oliguric and dismissing a rising creatinine with normal urine output — aminoglycoside and contrast ATN are characteristically non-oliguric, so normal urine output does NOT exclude significant tubular injury.

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8. Cystatin C

WHAT YOU SEE

CYSTATIN C, 'earlier than Cr', false high with steroids/hyperthyroid

WHAT IT ENCODES

Cystatin C is freely filtered and not secreted, so it rises EARLIER than creatinine and is independent of muscle mass — useful in cachectic, elderly, or amputee patients where creatinine underestimates injury. It is falsely elevated by corticosteroids and hyperthyroidism and falsely low in hypothyroidism, limiting it in those states.

BOARD TRAP

Dismissing early AKI because serum creatinine is still 'normal' — creatinine lags injury by 24–48h due to its reservoir, so a normal Cr does not rule out an acute GFR drop that cystatin C would already show.

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9. Kidney size US

WHAT YOU SEE

KIDNEY-SIZE COMPARISON: AKI normal, CKD shrunken, enlarged

WHAT IT ENCODES

On ultrasound, kidney size separates acute from chronic: NORMAL size and cortical thickness favor AKI (potentially reversible); SMALL, echogenic kidneys with thinned cortex indicate CKD (chronic, irreversible scarring). ENLARGED kidneys suggest infiltration (lymphoma, amyloid), HIV nephropathy, diabetic nephropathy, polycystic disease, or acute obstruction with hydronephrosis.

BOARD TRAP

Reading small echogenic kidneys as 'severe AKI that needs aggressive treatment' — small kidneys mean established CKD, which is irreversible, and the appropriate move is to recognize chronicity rather than pursue reversal.

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10. Biopsy indication

WHAT YOU SEE

BIOPSY microscope/needle, 'UO under 200 high risk'

WHAT IT ENCODES

Kidney biopsy is reserved for intrinsic AKI of unclear cause, suspected rapidly progressive glomerulonephritis/vasculitis, or AKI failing to recover when prerenal and postrenal are excluded — it changes management when immunosuppression is being considered. It is risk-stratified: bleeding risk rises with uncontrolled hypertension, single kidney, coagulopathy, and very low urine output.

BOARD TRAP

Ordering a biopsy as a first-line test — the trap is skipping non-invasive classification (ultrasound, sediment, serologies) and going straight to an invasive bleeding-prone procedure when the diagnosis is usually clinical.

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11. Urine sediment

WHAT YOU SEE

Microscope with cast tiers: pigmented, RBC, WBC, granular

WHAT IT ENCODES

Urine sediment is the single most discriminating bedside test. Muddy-brown granular ('dirty') casts = ATN; RBC casts with dysmorphic RBCs = glomerulonephritis; WBC casts = acute interstitial nephritis or pyelonephritis; broad waxy casts = CKD; bland sediment = prerenal or postrenal. Mnemonic: Muddy=ATN, Red=GN, White=AIN.

BOARD TRAP

Calling muddy-brown casts 'normal' or attributing them to prerenal disease — granular muddy casts are pathognomonic for ATN (intrinsic injury), whereas true prerenal urine is bland with no casts.

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12. Blood lab / CBC

WHAT YOU SEE

BLOOD-LAB CBC: schistocytes, eosinophils, low complement, TMA

WHAT IT ENCODES

Blood work narrows intrinsic causes: schistocytes + thrombocytopenia + high LDH/low haptoglobin = thrombotic microangiopathy (HUS/TTP); peripheral eosinophilia = AIN or atheroembolic disease (post-catheterization, also low complement + livedo); low C3/C4 = immune-complex GN (lupus, post-strep, MPGN, endocarditis); positive ANCA/anti-GBM = pauci-immune or anti-GBM RPGN.

BOARD TRAP

Attributing AKI with schistocytes and low platelets to 'DIC' and ordering plasma — in a young patient with normal coags this is TMA (TTP/HUS); the discriminator is normal PT/PTT/fibrinogen in TMA versus abnormal in DIC, and TTP needs urgent plasma exchange.

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13. GN / vasculitis

WHAT YOU SEE

GN/VASCULITIS label at bottom right

WHAT IT ENCODES

Glomerulonephritis/vasculitis presents as a nephritic picture: RBC casts, dysmorphic RBCs, sub-nephrotic-to-nephrotic proteinuria, hypertension, and edema. Serologic workup includes ANCA (granulomatosis with polyangiitis, microscopic polyangiitis), anti-GBM (Goodpasture), ANA/anti-dsDNA (lupus), complement levels, ASO/anti-DNase B, and cryoglobulins/hepatitis serologies. Rapidly progressive GN (crescents) is a nephrologic emergency needing urgent biopsy and immunosuppression.

BOARD TRAP

Mistaking nephritic RBC casts for a UTI or a 'traumatic' hematuria and missing RPGN — the discriminator is dysmorphic RBCs and RBC casts (glomerular origin) versus isomorphic RBCs and clots (urologic), and delay in RPGN costs nephrons irreversibly.
